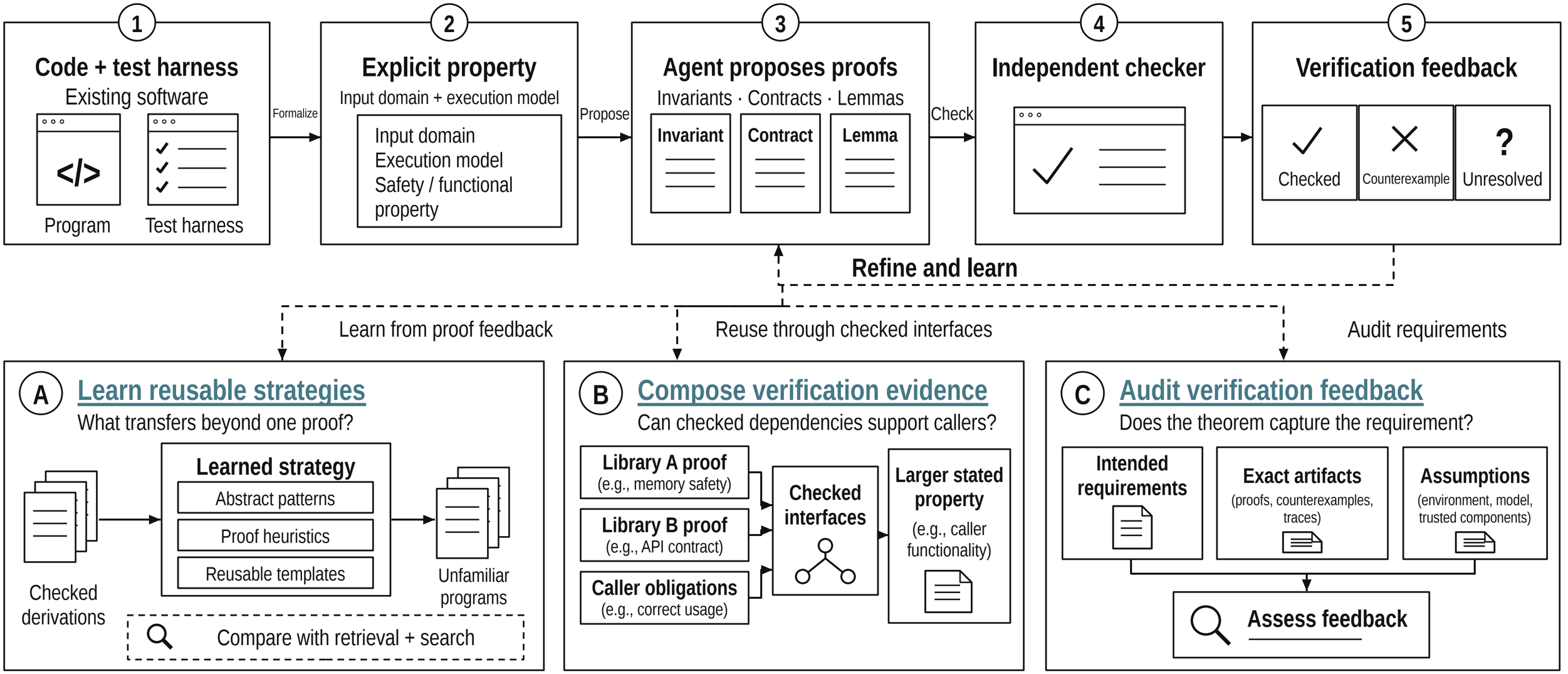
Learning Reusable Formal Reasoning

Proposed research

From expensive proofs to transferable reasoning



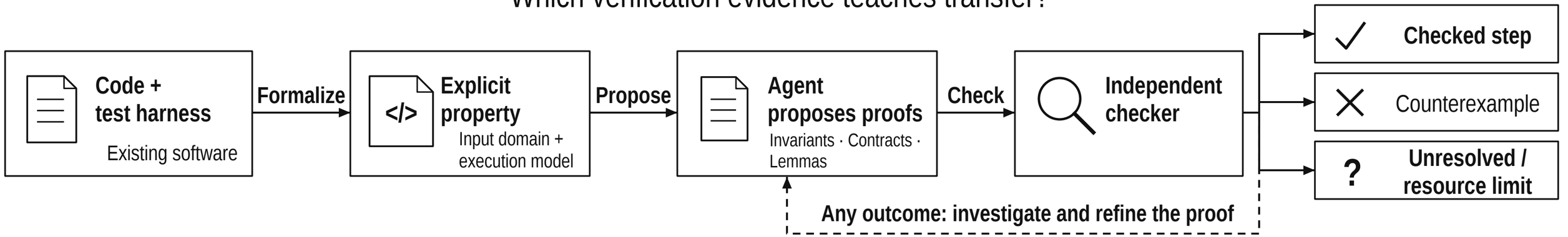
Efficient, reusable, independently checkable verification

Guarantees depend on the property, input domain, execution semantics, and trusted checking.

Learning Reusable Proof Strategies

Proposed research

Which verification evidence teaches transfer?



Compare three experimental mechanisms

1 Retrieve

Reuse checked proof artifacts

Search and reuse from
existing invariants, lemmas, etc.

2 Refine

Adapt during one task

Iteratively improve using
checker feedback on the same task.

3 Learn

Train reusable proof strategies

Strategy

Compare final proofs,
derivations, and revisions.

Held-out transfer test

Unfamiliar programs + changed obligations

Matched data, tools, and total cost

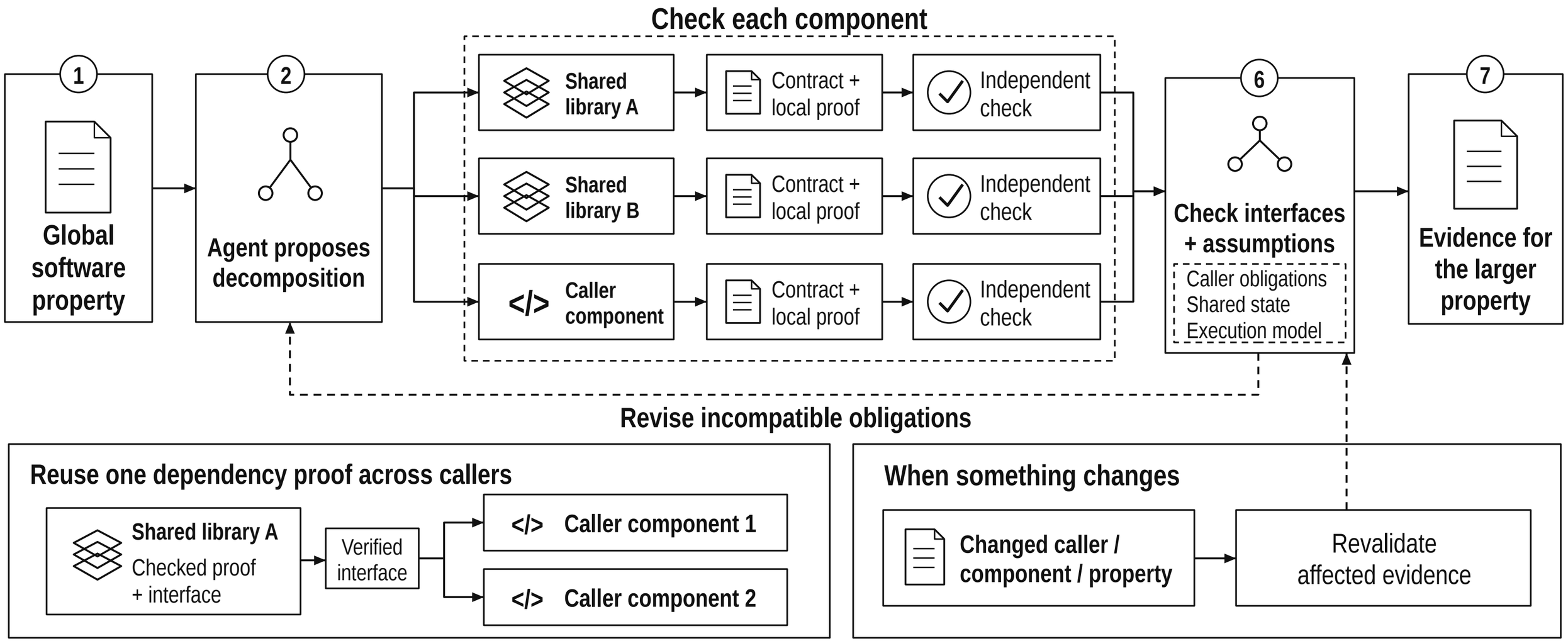
Sharing a proof and learning a strategy are evaluated separately.

Checked success does not establish generalization.

Compositional Verification

Proposed research

Reuse checked components through verified interfaces



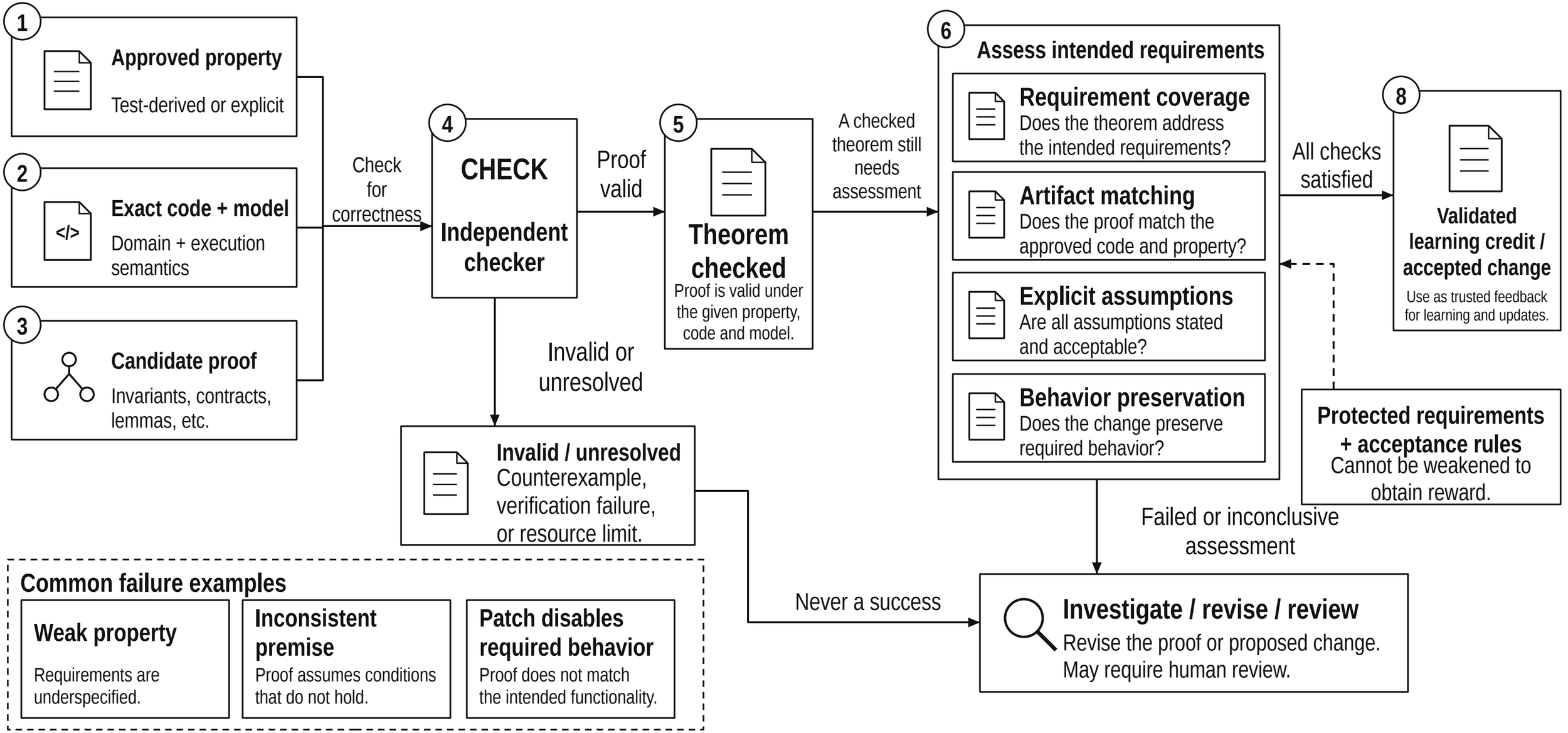
Reuse is conditional on matching interfaces, artifacts, and assumptions.

Larger guarantees require checked interfaces and explicit assumptions.

Trustworthy Verification Feedback

Proposed research

A checked theorem must support the intended requirement



A valid proof establishes its theorem; intended requirements need separate assessment.